Project Design Phase-II

Technology Stack (Architecture & Stack)

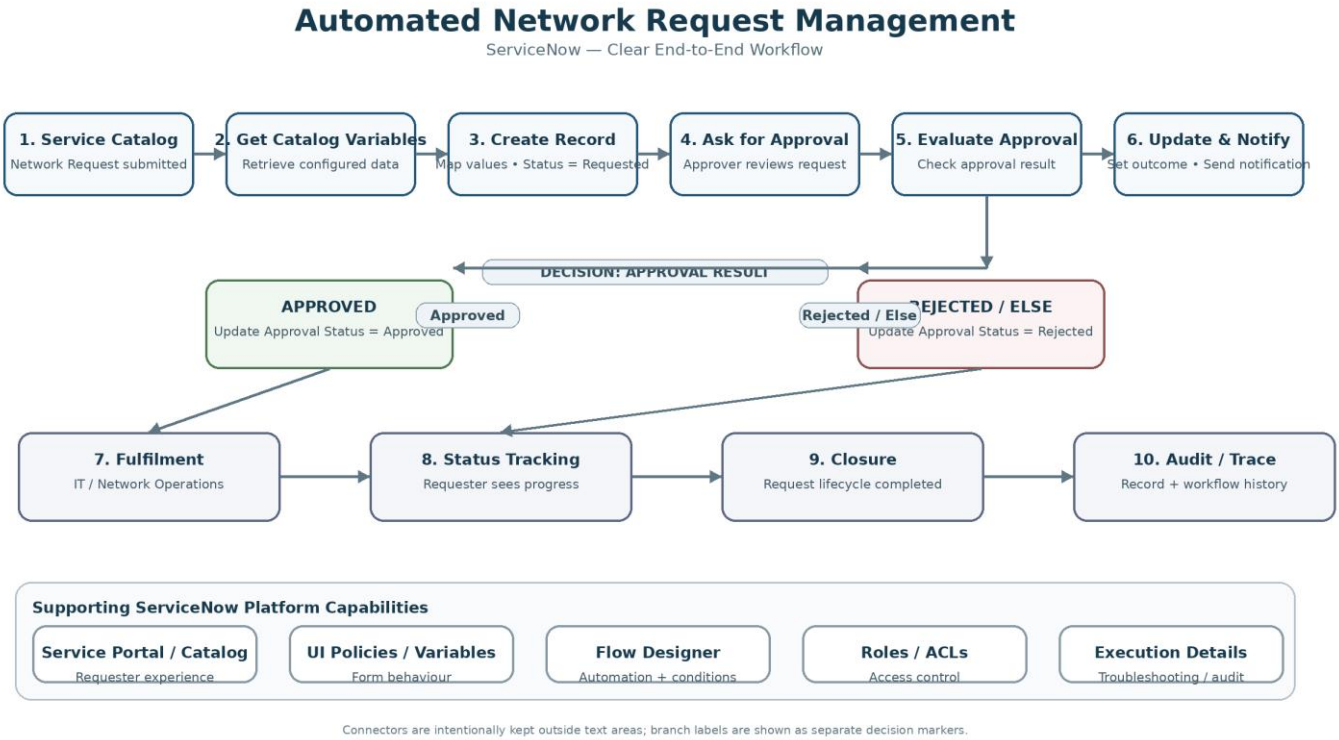
Date	14 October 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The architecture is centered on the ServiceNow cloud platform. The requester accesses the Network Request service through Service Portal and Service Catalog. Catalog variables and configured form behaviour capture the required request information. Flow Designer receives the submitted Requested Item context, retrieves the catalog variables, creates the configured operational record, sends the request for approval, evaluates the approval result, updates the record and sends lifecycle notifications. Roles and ACLs provide access control, while Flow Designer execution details and platform debugging outputs support troubleshooting.

Architecture / Workflow Diagram

Revised for clear separation of workflow stages, decision branches and supporting ServiceNow capabilities.



Infrastructure Demarcation

• Cloud / Platform: ServiceNow cloud instance (Personal Developer Instance during development/training).

• Client side: Web browser accessing Service Portal; no separate native mobile application is required for the documented solution.

• Application logic: ServiceNow configuration and low-code automation, primarily Service Catalog, UI Policies, catalog variables and Flow Designer.

• Data storage: ServiceNow Request / Requested Item records plus the configured operational record used by the current implementation.

• External interfaces: No production network-device API, payment gateway, SMS, WhatsApp or third-party integration is claimed as implemented in the current version.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Requester-facing submission and request-status experience.	ServiceNow Service Portal; Service Catalog; catalog variables
2	Form Behaviour	Guided form experience, visibility and mandatory-field behaviour.	ServiceNow UI Policies; Catalog Client Scripts where required
3	Request Data Capture	Structured network-request information used by downstream automation.	Service Catalog variables; mandatory fields; configured catalog-item structure
4	Workflow Automation	End-to-end orchestration of request processing, record creation, approval and branching.	ServiceNow Flow Designer; triggers; actions; conditions
5	Approval Management	Approval request, decision capture and approved/rejected branching.	Flow Designer Ask for Approval; approval rules; approval status condition
6	Operational Data	Structured record created and updated from submitted catalog values.	ServiceNow configured operational table / record; current implementation visibly uses the configured Database Tables record
7	Platform Records	Native request context generated from Service Catalog submission.	ServiceNow Request / Requested Item (REQ / RITM) records
8	Notifications	Request-created and resolution/status communications.	ServiceNow Flow Designer Send Email action / configured notifications
9	Security	Controlled access to request and operational information.	ServiceNow Roles; Access Control Lists (ACLs)
10	Diagnostics	Trace flow execution, mappings, conditions and failures.	Flow Designer Execution Details; platform logs / debugging outputs
11	File Storage	No separate application file-storage service is required by the documented core flow.	ServiceNow-managed platform storage as applicable
12	External API / Integration	Future integration point for network systems or external notification/services.	Not implemented in current version; integration can be added as future scope
13	Machine Learning Model	No ML inference or model dependency is required by the current workflow.	Not applicable to the current version
14	Infrastructure (Server / Cloud)	Application execution and hosting environment.	ServiceNow cloud instance / Personal Developer Instance for development and training

Technology Flow

Browser → Service Portal → Network Request Catalog Item → Catalog Variables / UI Policies → Flow Designer → REQ/RITM + Configured Operational Record → Approval → Record Update → Notification → IT / Network Operations.

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology / Design Choice
1	Open-Source Frameworks	No separate open-source application framework is required by the documented core implementation.	ServiceNow platform components; no specific external framework asserted
2	Security	Protect network-request records and administrative capabilities according to user role; enforce controlled read/write access.	ServiceNow roles, ACLs and platform authentication/session controls
3	Scalable Architecture	Configuration-driven design permits additional request categories, variables, approval rules and fulfilment paths without replacing the complete solution.	ServiceNow modular Service Catalog + Flow Designer + configured record design
4	Availability	Application availability follows the availability of the deployed ServiceNow instance and related platform services.	ServiceNow-managed cloud platform; no separate availability layer
5	Performance	Minimize unnecessary client scripting and use Flow Designer for maintainable server-side orchestration; validate variable mapping and workflow execution for normal request scenarios.	ServiceNow workflow execution + lightweight portal/client configuration
6	Maintainability	Use configuration-first components and document the setup so another administrator	Service Catalog, Flow Designer, UI Policies, ACLs and documented configuration

		can reproduce or modify the solution.	
7	Observability / Troubleshooting	Trace workflow runs, action inputs/outputs, condition paths and record mappings when failures occur.	Flow Designer Execution Details; platform logs/debug outputs
8	Compatibility	Ensure the requester experience operates through the intended Service Portal/catalog experience and does not depend on backend-only UI behaviour.	Service Portal + Service Catalog + supported ServiceNow configuration

Implemented vs Future Technology Boundary

• Implemented/documented core: Service Portal, Service Catalog, catalog variables, configured form behaviour, Flow Designer, operational record creation/update, approval handling, condition branching, email notifications, roles/ACLs and execution tracing.

• Future integrations: network-management/device APIs, automated fulfilment integrations, external messaging channels and advanced operational dashboards.

• Not applicable in the current version: a machine-learning inference model and a separately managed application server/database outside ServiceNow.